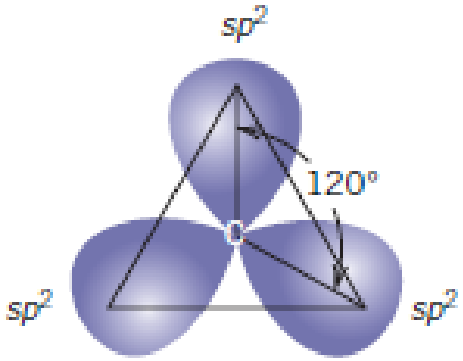
Introduction and Structures of Materials - IV

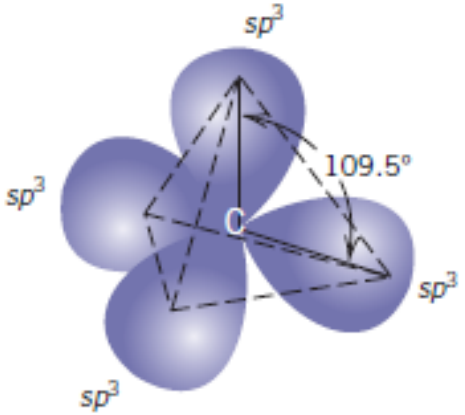


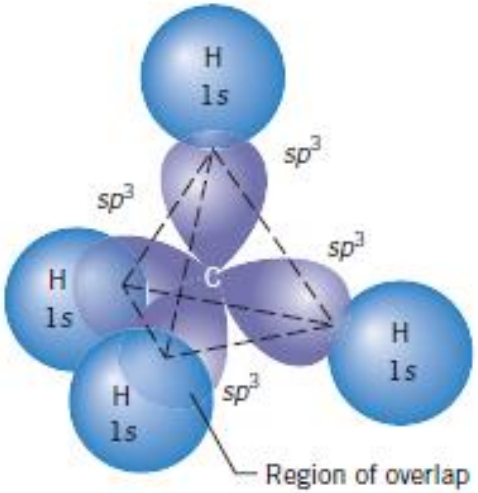
MO-201 Materials Science
Odd Sem 2026-27

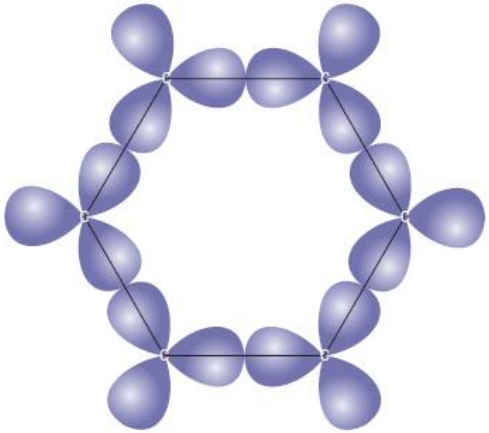
Dr. S. Sinha, Dept. of Metallurgical Engg.
IIT (BHU) Varanasi

Covalent Solids

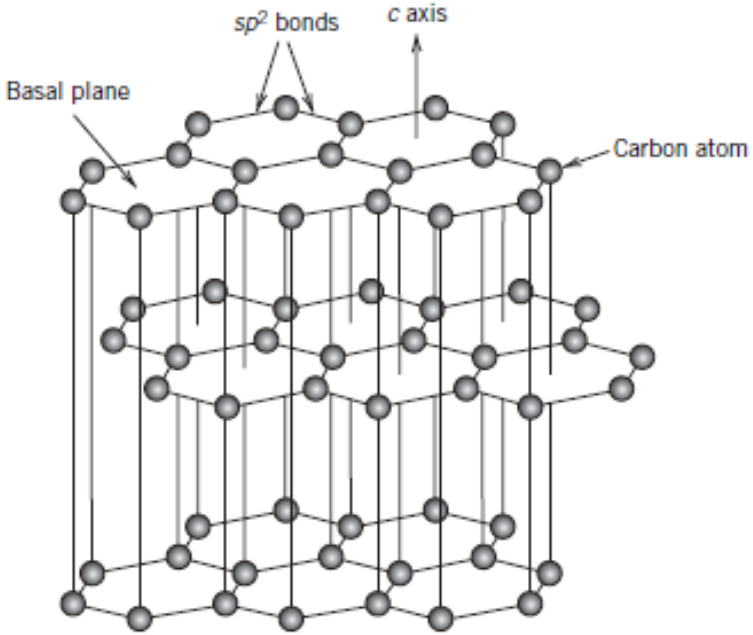


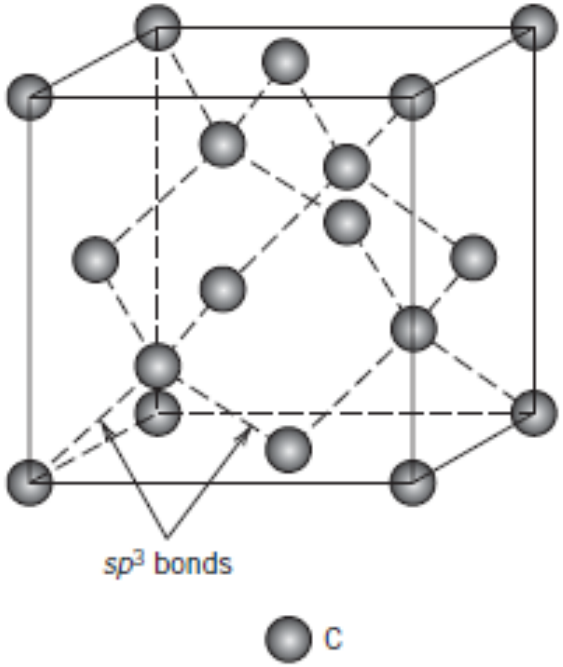






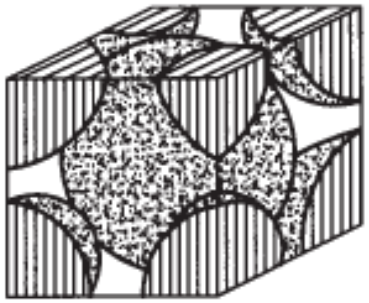
The formation of a hexagon by the bonding of 6 sp^2 triangles to one another





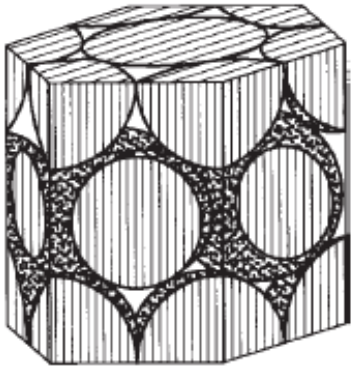
Metals and Alloys





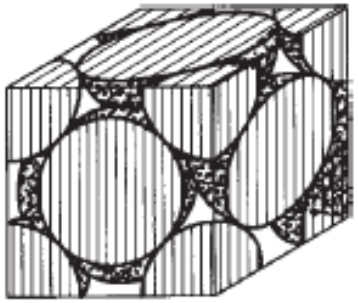
BCC unit cell





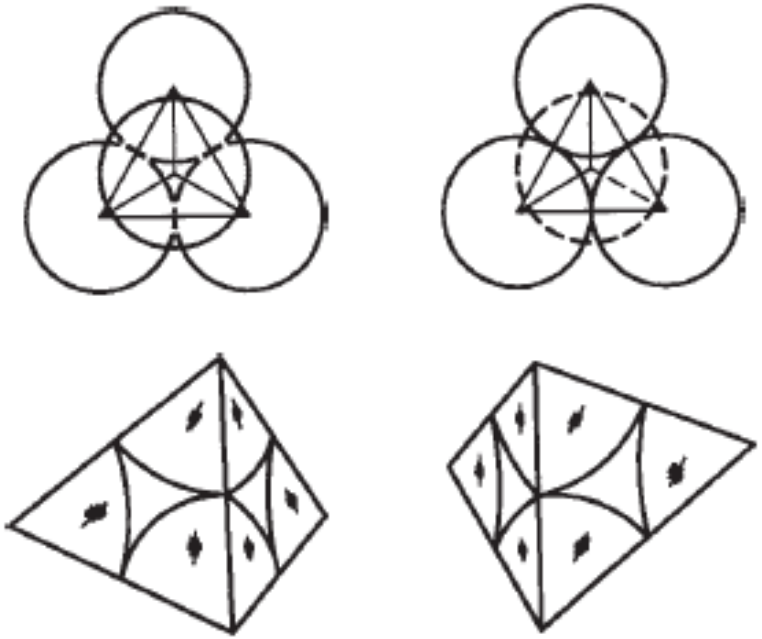
HCP unit cell





FCC unit cell

Voids

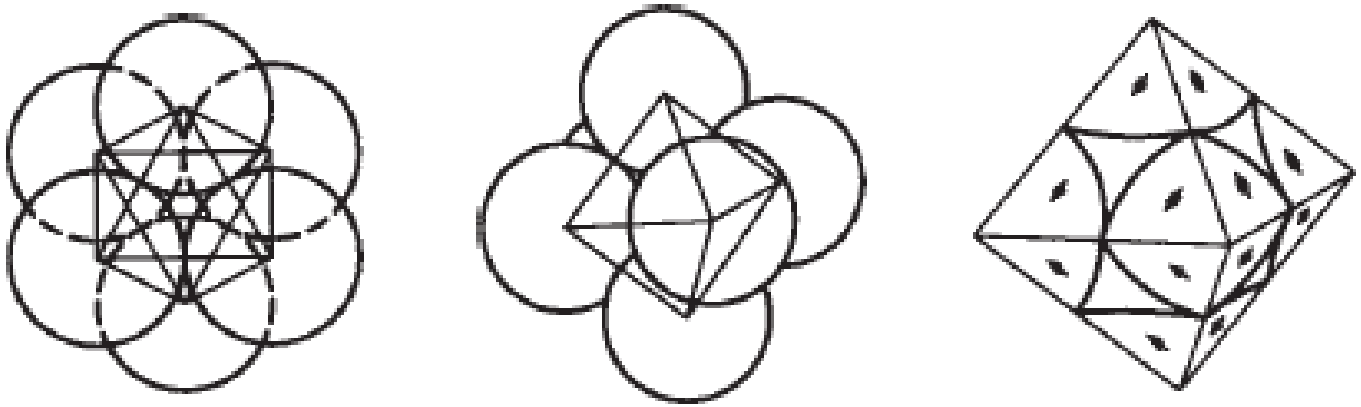


Tetrahedral voids

r = 0.225R

Octahedral voids

r = 0.414R



Voids

Derivation of size of tetrahedral void

- Let the edge length of the cubic unit cell be a, the radius of the large host sphere be R , and the radius of the small sphere inside the void be r.
- The distance between the corner atom and the center of the tetrahedral void along the body diagonal is $\frac{\sqrt{3}a}{4}$.
- At the same time, this distance is equal to the sum of the radii of the corner atom and the void atom:

$$\frac{\sqrt{3}\,a}{4} = R + r$$

- In an FCC lattice, the face diagonal relates the edge length a to the sphere radius R via:

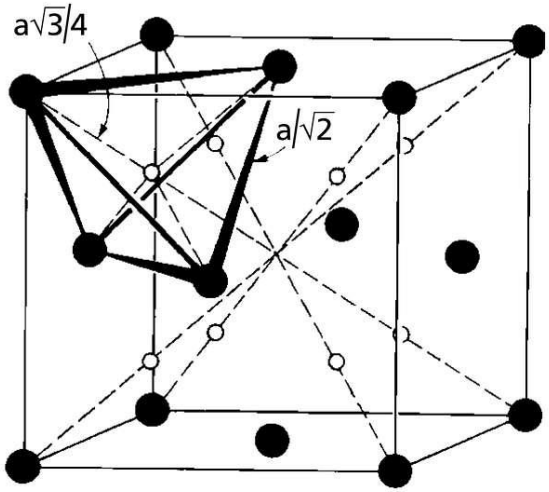
$$\sqrt{2}\,a = 4R \Longrightarrow a = \frac{4R}{\sqrt{2}} = 2\sqrt{2}\,R$$

Voids

Derivation of size of tetrahedral void

- Substitute this expression for a into the tetrahedral distance equation:

$$\frac{\sqrt{3}(2\sqrt{2}\,R)}{4} = R + r$$



- Simplify the fraction on the left side:

$$\frac{2\sqrt{6}\,R}{4} = R + r \Longrightarrow \frac{\sqrt{6}}{2} R = R + r$$

- Substitute the numerical value of $\sqrt{6} \approx 2.449$:

$$1.225R = R + r$$

- Subtract R from both sides to isolate r:

$$r = 1.225R - R = 0.225R$$

- Thus, the ratio $\frac{r}{R} = 0.225$.

Voids

Derivation of size of octahedral void

- Connect the centers of the adjacent spheres to form a square with side length equal to 2R (since adjacent spheres touch each other, the distance between their centers is R + R = 2R).
- The diagonal (d) of this square passes through the center of the void and connects the centers of diagonally opposite spheres.
- The length of this diagonal in terms of the side length (a = 2R) using the Pythagorean theorem is:

$$\text{Diagonal} = \sqrt{a^2 + a^2} = a\sqrt{2} = 2R\sqrt{2}$$

- Alternatively, the diagonal length is expressed by adding the radii of the two outer spheres and the diameter of the void space:

$$\text{Diagonal} = R + r + r + R = 2R + 2r = 2(R + r)$$

Voids

Derivation of size of octahedral void

- Set the two expressions for the diagonal equal to each other:

$$2(R + r) = 2R\sqrt{2}$$

- Divide both sides by 2:

$$R + r = R\sqrt{2}$$

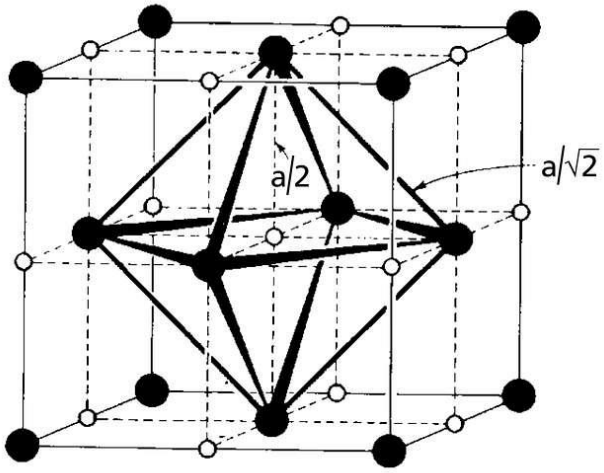
- Substitute the numerical value of $\sqrt{2} \approx 1.414$:

$$R + r = 1.414R$$

- Isolate r:

$$r = 1.414R - R = 0.414R$$

- Thus, r = 0.414R.



Solid Solutions

- When a solute atom is much smaller than the solvent atom, it may dissolve interstitially occupying a void space in the parent structure

- For example, carbon is an interstitial solute in FCC iron and occupies the octahedral voids in the FCC structure

- When the solute and the solvent atoms are of comparable sizes, the solute substitutes for the solvent atom on a regular atomic site

- For example, a 70% Cu-30% Zn alloy (alpha brass) has an FCC structure, with copper and zinc atoms occupying randomly the atomic sites of the FCC crystal

- Hume Rothery has framed empirical rules that govern the formation of substitutional solid solutions

Solid Solutions

Parameters Relevant to the Hume Rothery Rules

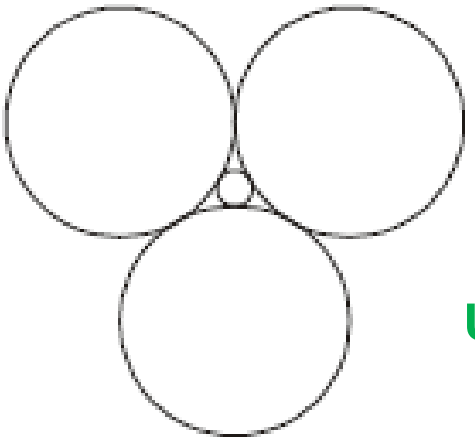
System		Crystal structure	Radius of atoms, Å	Valency	Electro-negativity
Ag–Au	Ag	FCC	1.44	1	1.9
	Au	FCC	1.44	1	2.4
Cu–Ni	Cu	FCC	1.28	1	1.9
	Ni	FCC	1.25	2	1.8
Ge–Si	Ge	DC	1.22	4	1.8
	Si	DC	1.18	4	1.8

1) Solute and the solvent atoms do not differ by more than 15% in diameter
2) Electronegativity difference between the elements is small
3) Valency and the Crystal Structure of the elements are the same

- This obviously is not possible in the case of Cu and Zn which have FCC and HCP structures, respectively
- They, therefore, form solid solutions up to a certain extent only
- At room temperature, up to 35% of Zn can dissolve in the FCC crystal of Cu, as the Hume Rothery conditions are partially satisfied

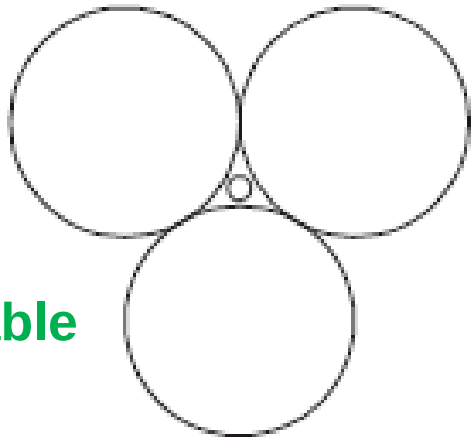
Ionic Solids

Critical

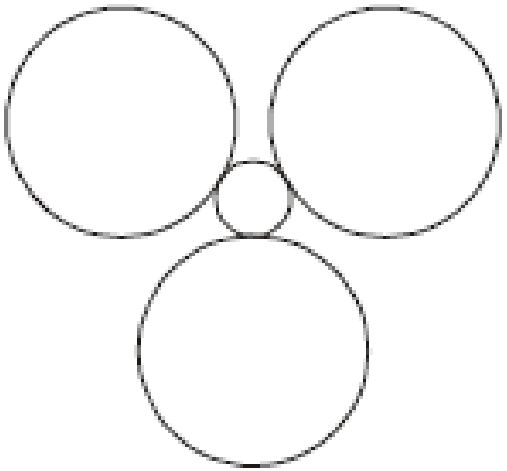


$r_c/r_a = 0.155$

Unstable



$0 < r_c/r_a < 0.155$



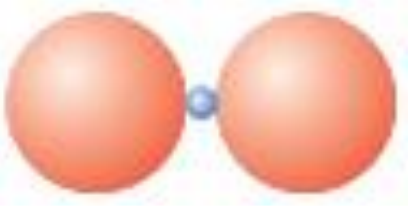
$0.155 < r_c/r_a < 0.225$

Stable (but not critical)

Ligancy as a Function of Radius Ratio

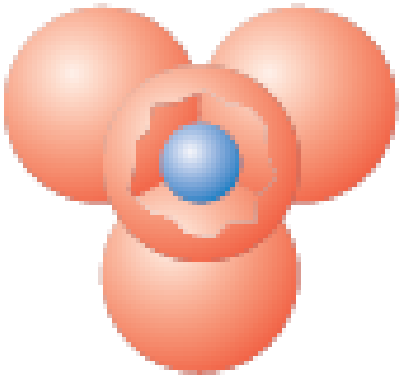
Ligancy	Range of radius ratio	Configuration
2	0.0 – 0.155	linear
3	0.155 – 0.225	triangular
4	0.225 – 0.414	tetrahedral
6	0.414 – 0.732	octahedral
8	0.732 – 1.0	cubic
12	1.0	"FCC" or "HCP"

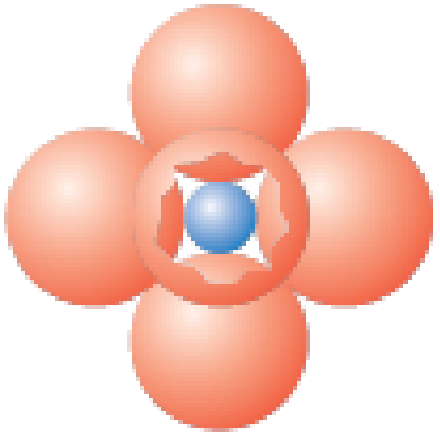
Ionic Solids



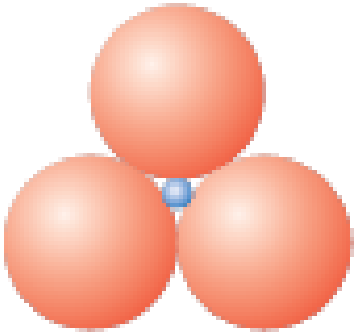
CN = 2
r_C/r_A <0.155

CN = 4
r_C/r_A = 0.225 – 0.414

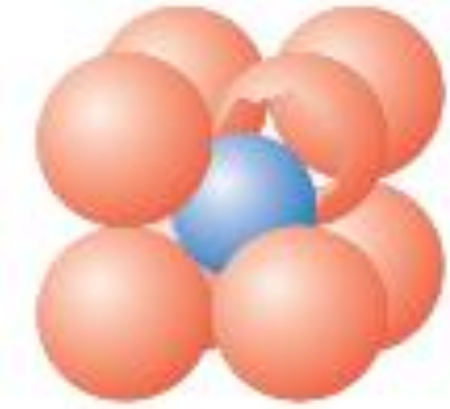




CN = 6
r_C/r_A = 0.414 – 0.732



CN = 3
r_C/r_A = 0.155 – 0.225



CN = 8
r_C/r_A = 0.732 – 1

Ionic Solids

- The ligancy rules outlined based on radius ratio are obeyed in a number of cases

- For example, in the NaCl crystal, the radius ratio r_{Na+}/r_{Cl-} = 0.54, which lies between 0.414 and 0.732; hence, the predicted ligancy is 6 and octahedral geometry of six chlorine ions surrounding a central cation is experimentally observed

- In MgO, r_{Mg2+}/r_{O2-} = 0.59 and again the octahedral coordination is observed

- In CsCl, r_{Cs+}/r_{Cl-} = 0.91, which lies in the range of 0.732–1.0; the predicted coordination of eight anions surrounding a cation is observed

- In a marginal case such as CaF_2, where r_{Ca2+}/r_{F-} = 0.73, it is difficult to predict whether a six-fold or an eight-fold coordination will occur; it so turns out that the eight coordination is observed in this case, that is, every calcium cation is surrounded by eight fluorine anions

Ionic Solids

- The Si–O bond in silica as well as in silicates is about 50% ionic and 50% covalent; here, the central silicon cation is surrounded by four oxygen anions located at the corners of a regular tetrahedron

- This arrangement satisfies both the ligancy rules (as $r_c/r_a = 0.29$, the tetrahedral coordination is predicted) as well as the orientation relationships of (sp^3) bonds

- The stability criteria listed above for predicting the ligancy may not always be valid; if directional characteristics of bonding persist to any significant degree, the considerations based on the radius ratio alone will not lead to the correct prediction of ligancy

- In the above-discussed example of Si–O coordination, the radius ratio criterion and the bond angle requirements happen to coincide; in ZnS, where the bond is more covalent than ionic, the ligancy predicted from $r_{Zn2+}/r_{S2-} = 0.48$ is octahedral; yet the four-fold coordination characteristic of (sp^3) bonding is what is observed

Ionic Solids

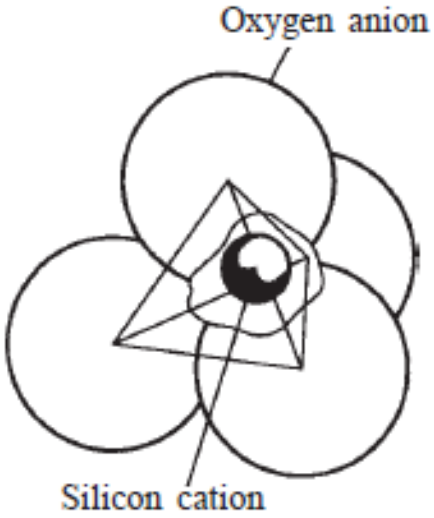
- In the formation of ionic crystals, the ligancy rules described above determine the local packing around a cation; the long-range arrangement of ions in the crystal is dependent on the following factors

1. In the crystal, the **overall electrical neutrality should be maintained**, whatever be the net charge on a local group of a cation and surrounding anions. For example, in NaCl, where a cation is surrounded by six anions, the net charge on ($NaCl_6$) is five. Evidently, this has to be neutralized in the long range arrangement.

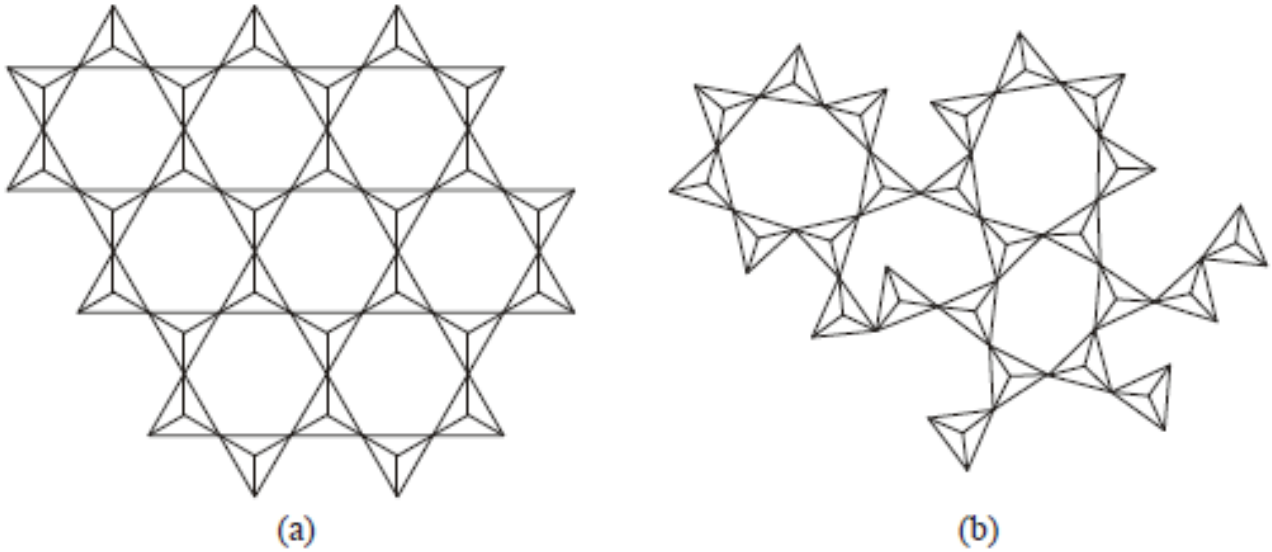
2. The ionic bond being nondirectional, the **ions are packed as closely as possible** in the crystal, **consistent with the local coordination**

3. If **small cations with a large net charge are present in the crystal, the cation–cation repulsion will be high**. Then it may be necessary to have a long range arrangement that maximizes the cation–cation distance, even if close packing is not possible. Such a situation usually arises when the charge on the cation increases to 3 or 4.

Structure of Silica and Silicates

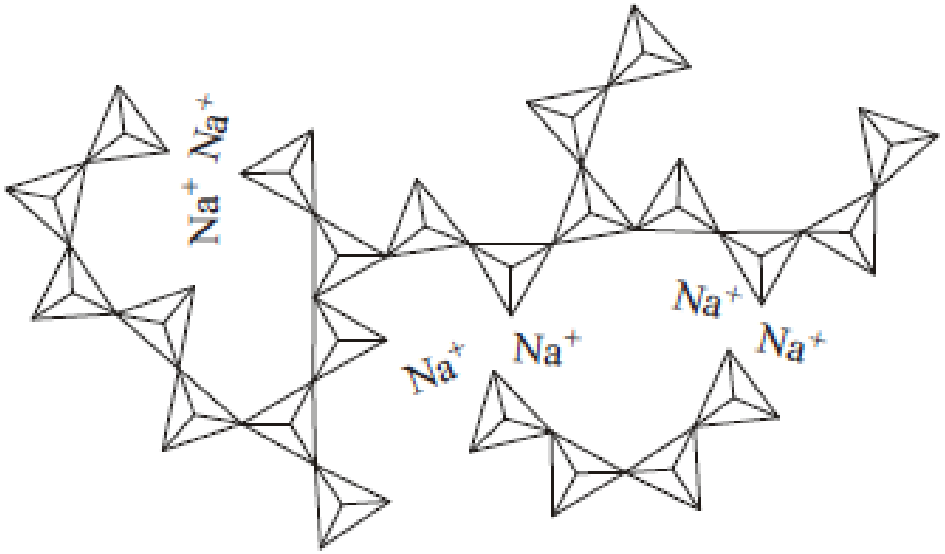


The silicate tetrahedron has a silicon cation at the center of the tetrahedron and four oxygen anions at the four corners



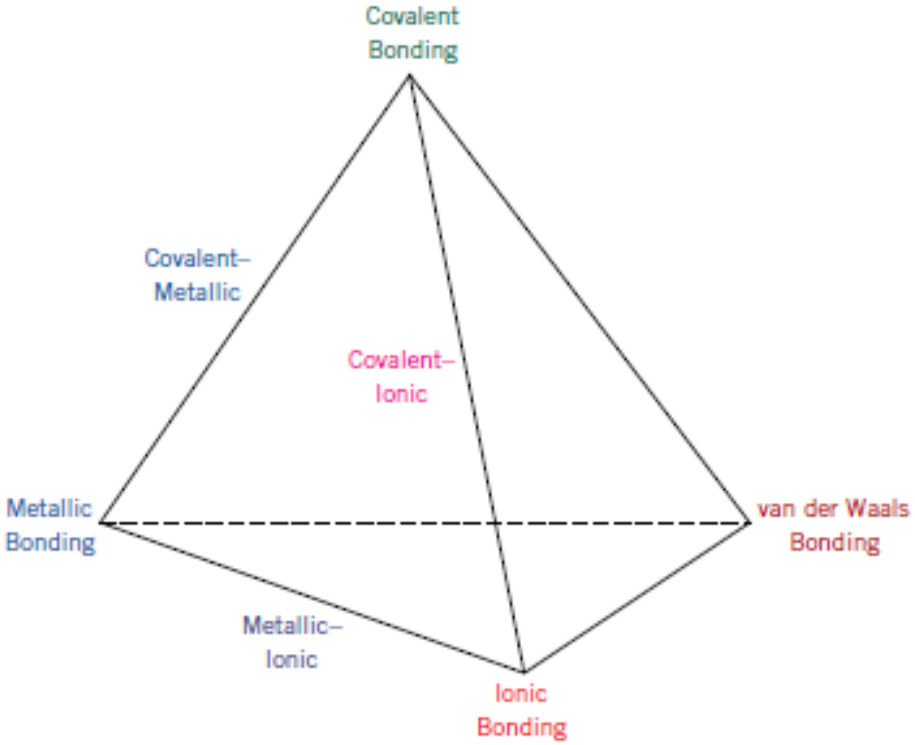
Structure of silica in (a) crystalline and (b) noncrystalline forms

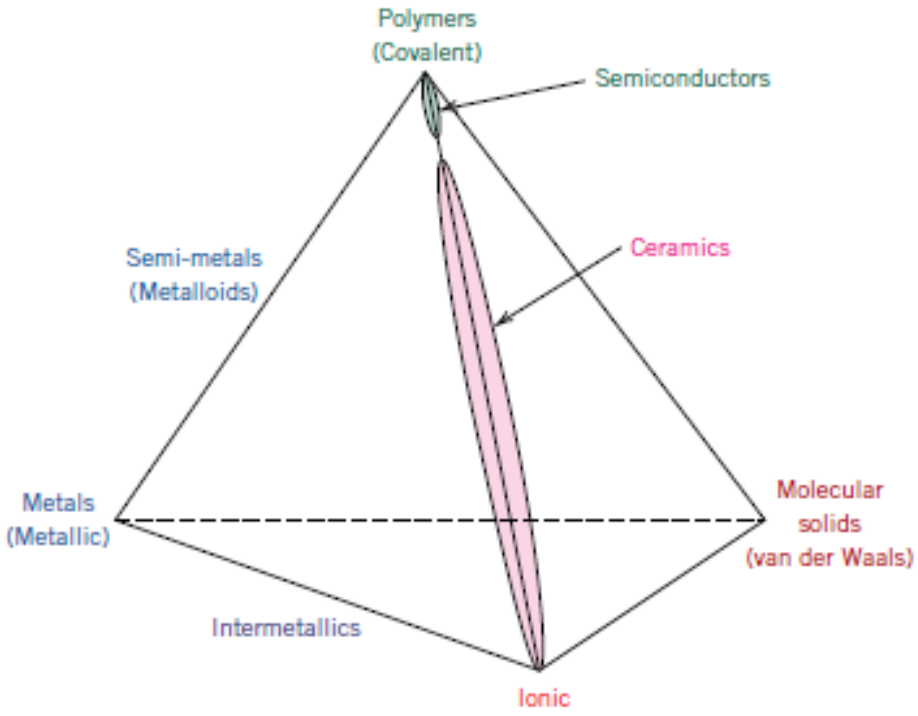
Structure of Silica and Silicates



Addition of Na_2O to silica introduces weaker bonds in the network

Mixed Bonding





Polymers

- Most organic polymers are based on the covalent bonds formed by carbon; the electrons are bonded strongly by the localized sharing, characteristic of covalent bonding so that polymers are good thermal and electrical insulators

- Relative inertness of polymeric materials (they do not corrode in the sense that metals do) can be attributed to covalent bonding

- The word ‘mer’ in Greek means a unit, and so monomer stands for a single unit and polymer for many units joined together; a polymer usually has thousands of monomers bonded by a chemical reaction; for this reason, it is often called a **macromolecule**

- If joining occurs essentially along one direction, we produce **long chain polymers**; these long chains are held together by secondary bonds in a polymeric solid; If joining occurs in other directions as well, we may produce a network of primary bonds yielding a **three-dimensional giant molecule**

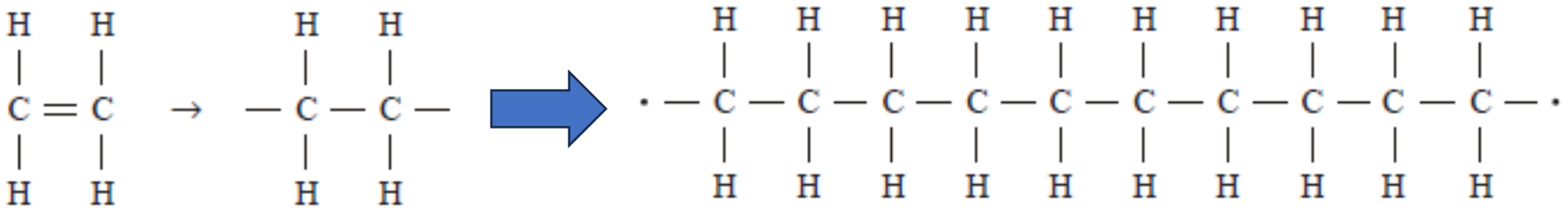
Polymers

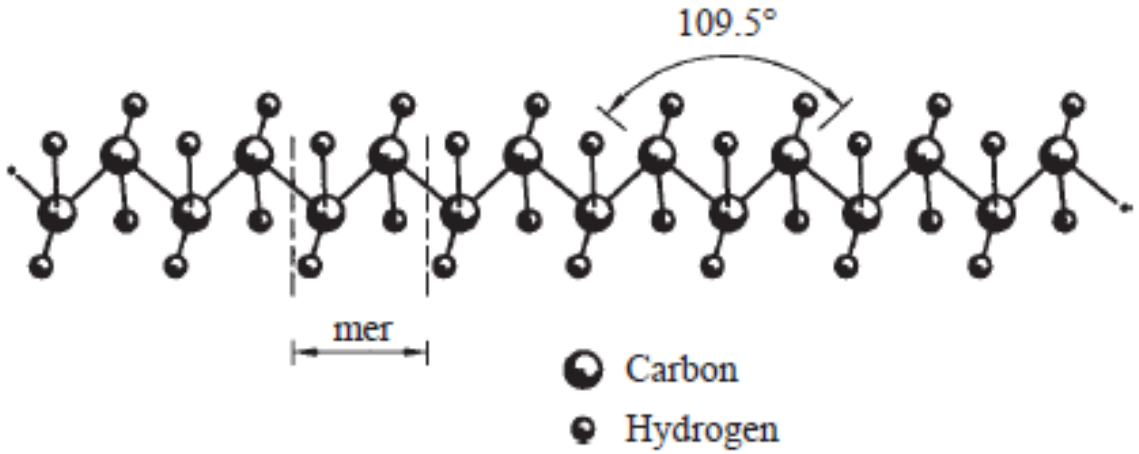
- Classified into **Thermoplasts** and **Thermosets** depending on their behaviour with a change in temperature
- Thermoplasts have the property of increasing plasticity, that is, increasing ability to deform plastically with increasing temperature
- Thermoplasts are long chain molecules held together by secondary bonds
- Thermosets have a three-dimensional network of primary bonds as polymerization proceeds in all directions; they are, therefore, relatively hard and rigid at room temperature and do not soften on heating; in fact, they become harder due to the completion of any left-over polymerization reaction
- However, on heating, before reaching the melting temperature, they degrade, that is, they decompose owing to their reaction with atmospheric oxygen
- Bakelite used for electrical insulation is a thermoset; the smell of Bakelite in a short circuited electrical appliance is a sign of degradation

Polymers

- Long chain polymers further subdivided into Plastics, Fibres and Elastomers

- The long chain molecules are randomly oriented in a plastic; in fibres, all the chains are more or less aligned in the long direction of the fibre; elastomers are long chain molecules exhibiting the unique rubbery behaviour



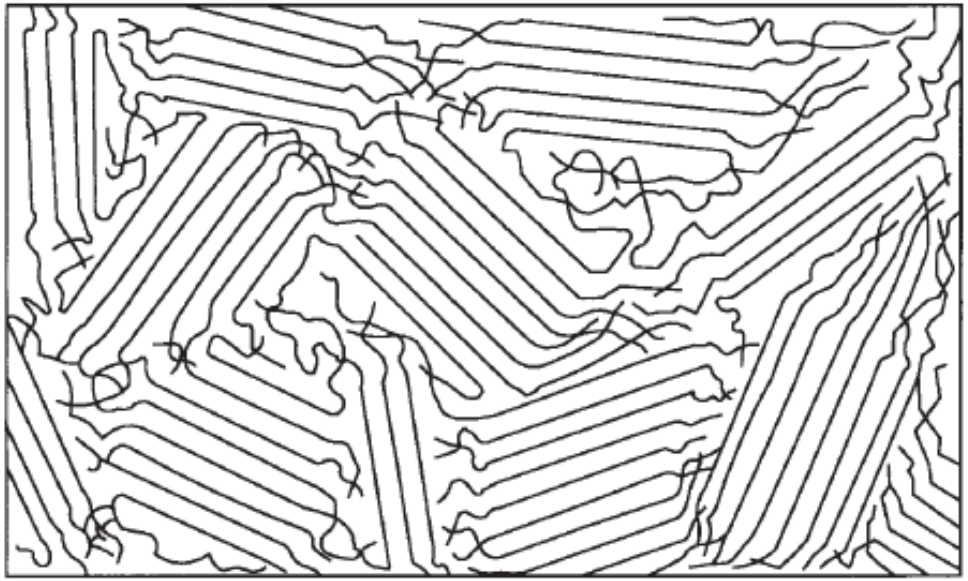


The structure of a polyethylene chain

Crystallinity of Long Chain Polymers

Long chain polymers are usually in the noncrystalline form or in the semicrystalline form. In the noncrystalline form, the long chains are randomly tangled with one another. In the semicrystalline form, parts of the polymer volume have the parallel chain arrangement while other parts are randomly oriented

In all polymers, the long chains can be aligned to some extent by mechanical working; such an alignment promotes crystallinity



Many polymers are semicrystalline, with random and parallel arrangements of chains in different regions

References

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- Fundamentals of Materials Science and Engineering, W.D. Callister